

```
!pip install tensorflow
```

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras import layers, models
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.19.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: google-pasta>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: typing-extensions>=3.6.6 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: tensorboard~2.19.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: keras>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: numpy<2.2.0,>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow)
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Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0)
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Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from rich)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from rich)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.19.0)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.19.0)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.19.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich)
Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0)
```

```
# Load MNIST dataset
(x_train, _), (x_test, _) = tf.keras.datasets.mnist.load_data()

# Normalize
x_train = x_train / 255.0
x_test = x_test / 255.0

# Reshape (add channel dimension)
x_train = np.reshape(x_train, (-1, 28, 28, 1))
x_test = np.reshape(x_test, (-1, 28, 28, 1))

print("Shape:", x_train.shape)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 — 0s 0us/step
Shape: (60000, 28, 28, 1)
```

```

noise_factor = 0.5

x_train_noisy = x_train + noise_factor * np.random.normal(size=x_train.shape)
x_test_noisy = x_test + noise_factor * np.random.normal(size=x_test.shape)

# Clip values between 0 and 1
x_train_noisy = np.clip(x_train_noisy, 0., 1.)
x_test_noisy = np.clip(x_test_noisy, 0., 1.)

```

```

input_img = layers.Input(shape=(28, 28, 1))

# Encoder
x = layers.Conv2D(32, (3,3), activation='relu', padding='same')(input_img)
x = layers.MaxPooling2D((2,2), padding='same')(x)
x = layers.Conv2D(16, (3,3), activation='relu', padding='same')(x)
encoded = layers.MaxPooling2D((2,2), padding='same')(x)

# Decoder
x = layers.Conv2D(16, (3,3), activation='relu', padding='same')(encoded)
x = layers.UpSampling2D((2,2))(x)
x = layers.Conv2D(32, (3,3), activation='relu', padding='same')(x)
x = layers.UpSampling2D((2,2))(x)
decoded = layers.Conv2D(1, (3,3), activation='sigmoid', padding='same')(x)

# Model
autoencoder = models.Model(input_img, decoded)
autoencoder.compile(optimizer='adam', loss='binary_crossentropy')

autoencoder.summary()

```

Model: "functional"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 28, 28, 1)	0
conv2d (Conv2D)	(None, 28, 28, 32)	320
max_pooling2d (MaxPooling2D)	(None, 14, 14, 32)	0
conv2d_1 (Conv2D)	(None, 14, 14, 16)	4,624
max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 16)	0
conv2d_2 (Conv2D)	(None, 7, 7, 16)	2,320
up_sampling2d (UpSampling2D)	(None, 14, 14, 16)	0
conv2d_3 (Conv2D)	(None, 14, 14, 32)	4,640
up_sampling2d_1 (UpSampling2D)	(None, 28, 28, 32)	0
conv2d_4 (Conv2D)	(None, 28, 28, 1)	289

Total params: 12,193 (47.63 KB)

```

history = autoencoder.fit(
    x_train_noisy, x_train,
    epochs=5,
    batch_size=128,

```

```
validation_data=(x_test_noisy, x_test)
)
```

```
Epoch 1/5
469/469 ————— 109s 227ms/step - loss: 0.1727 - val_loss: 0.1212
Epoch 2/5
469/469 ————— 97s 206ms/step - loss: 0.1169 - val_loss: 0.1140
Epoch 3/5
469/469 ————— 140s 201ms/step - loss: 0.1112 - val_loss: 0.1085
Epoch 4/5
469/469 ————— 94s 201ms/step - loss: 0.1084 - val_loss: 0.1064
Epoch 5/5
469/469 ————— 95s 202ms/step - loss: 0.1064 - val_loss: 0.1050
```

```
decoded_imgs = autoencoder.predict(x_test_noisy)

n = 10
plt.figure(figsize=(20,6))

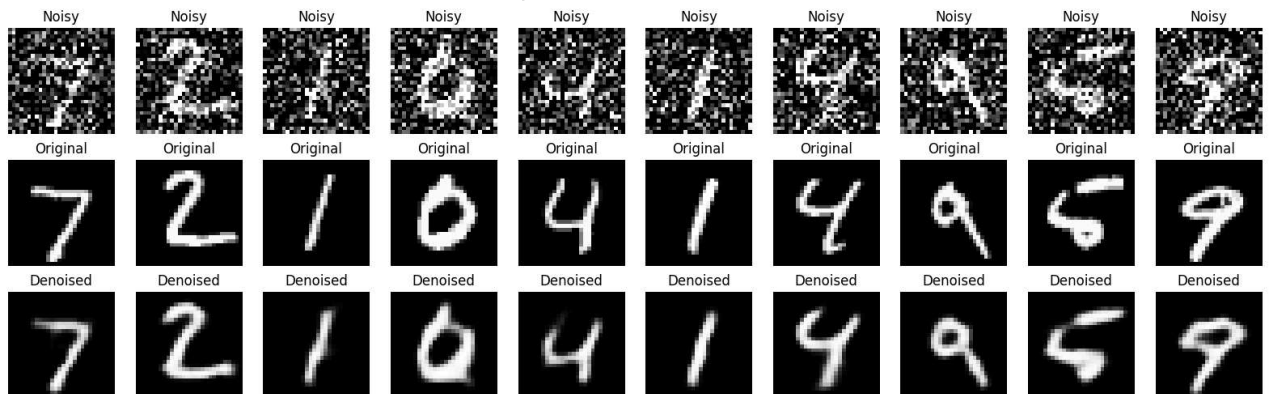
for i in range(n):
    # Noisy
    ax = plt.subplot(3, n, i+1)
    plt.imshow(x_test_noisy[i].reshape(28,28), cmap='gray')
    plt.title("Noisy")
    plt.axis("off")

    # Original
    ax = plt.subplot(3, n, i+n+1)
    plt.imshow(x_test[i].reshape(28,28), cmap='gray')
    plt.title("Original")
    plt.axis("off")

    # Denoised
    ax = plt.subplot(3, n, i+2*n+1)
    plt.imshow(decoded_imgs[i].reshape(28,28), cmap='gray')
    plt.title("Denoised")
    plt.axis("off")

plt.show()
```

313/313 ————— 6s 19ms/step



```
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.legend()
plt.title("Loss Curve")
plt.show()
```

